# T3 UV one-loop RGE term comparison

## Notation key

| symbol | interaction / definition |
|---|---|
| $g_1$ | $D_\mu \supset i g_1 Y B_\mu$ |
| $g_2$ | $D_\mu \supset i g_2 T^A W_\mu^A$ |
| $g_3$ | $D_\mu \supset i g_3 t^A G_\mu^A$ |
| $Y_u$ | $-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$ |
| $Y_d$ | $-\bar{Q} Y_d H d + \text{h.c.}$ |
| $Y_e$ | $-\bar{L} Y_e H e + \text{h.c.}$ |
| $y_1$ | $L F S_1 + \text{h.c.}$ |
| $y_2$ | $L F S_2 + \text{h.c.}$ |
| $M_F$ | $-\bar{F} M_F F$ |
| $T$ | $\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$ | $\text{Tr}(y_1^\dagger y_1)$ |
| $T_\nu^{(2)}$ | $\text{Tr}(y_2^\dagger y_2)$ |
| $T_\nu$ | $T_\nu^{(1)} + T_\nu^{(2)}$ |
| $m_1^2$ | $m_1^2 S_1^\dagger S_1$ |
| $m_2^2$ | $m_2^2 S_2^\dagger S_2$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$ |
| $\lambda_{S_1}^{(1)}$ | $\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$ |
| $\lambda_{S_2}^{(1)}$ | $\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$ |
| $\lambda_{HS_1}^{(1)}$ | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$ |
| $\lambda_{HS_2}^{(1)}$ | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$ |
| $\lambda_{HS_1}^{(A)}$ | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$ |
| $\lambda_{HS_2}^{(A)}$ | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$ |
| $\lambda_5$ | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$ |
| $\lambda_{12}^{(1)}$ | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$ |
| $\lambda_{12}^{(A)}$ | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$ |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$ |
| $\lambda_{S_1}^{(A)}$ | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$ |
| $\lambda_{S_2}^{(A)}$ | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$ |
| $\alpha$ | $Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$ |
| $C_2(d)$ | $(d^2-1)/4$ |
| $T(d)$ | $d(d^2-1)/12$ |
| $C_{LLS_1S_2}^{ij;AB}$ | $L_i L_j S_1^A S_2^B$ |
| $C_{ijab}$ | $L_i L_j \phi_a \phi_b$ |
| $C_5^{ij}$ | $(L_i^T C L_j) H H$ |

| | |
|---|---|
| $R_W$ | $16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$ |

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \to g$ conversion as in the previous report.

$\beta_{g_1}^{(1)}$

**Quantity:** $g_1$

$$16\pi^2\,\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{41{g_1}^3}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\frac{41{g_1}^3}{2}$ |

$\beta_{g_2}^{(1)}$

**Quantity:** $g_2$

$$16\pi^2\,\mu\frac{dg_2}{d\mu} = \beta_{g_2}^{(1)}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{31{g_2}^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\frac{31{g_2}^3}{6}$ |

$\beta_{g_3}^{(1)}$

**Quantity:** $g_3$

$$16\pi^2\,\mu\frac{dg_3}{d\mu} = \beta_{g_3}^{(1)}$$

| Column | Representative term structure |
|---|---|
| T1 | $-7{g_3}^3$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-7{g_3}^3$ |

$\beta^{(1)}_{Y_u}$

**Quantity:** $Y_u$

$$16\pi^2\,\mu\frac{dY_u}{d\mu} = \beta^{(1)}_{Y_u}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ |
| T2 | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ |
| T3 | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |

**Quantity:** $Y_u$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ |
| T2 | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ |
| T3 | $-8{g_3}^2\ (Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |

**Quantity:** $Y_u$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |

$\beta^{(1)}_{Y_d}$

**Quantity:** $Y_d$

$$16\pi^2\,\mu\frac{dY_d}{d\mu} = \beta^{(1)}_{Y_d}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ |
| T2 | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ |
| T3 | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |

**Quantity:** $Y_d$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ |
| T2 | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ |
| T3 | $-8{g_3}^2\ (Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |

**Quantity:** $Y_d$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |

$\beta_{Y_e}^{(1)}$

**Quantity:** $Y_e$

$$16\pi^2\,\mu\frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ |
| T2 | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ |
| T3 | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |

**Quantity:** $Y_e$

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T2 | $-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$ |
| T3 | $-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$ |

$\beta_{y_1}^{(1)}$

**Quantity:** $y_1$

$$16\pi^2 \, \mu \frac{dy_1}{d\mu} = \beta_{y_1}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{3}{2} T_\nu^{(1)} \, (y_1)_{ij}$ |
| T2 | $+\frac{1}{4}(3 \left(y_2 \, (y_2)^\dagger \, y_1\right)_{ij} + 2 \left(Y_e \, (Y_e)^\dagger \, y_1\right)_{ij} + 5 \left(y_1 \, (y_1)^\dagger \, y_1\right)_{ij})$ |
| T3 | $-\frac{33}{4} {g_2}^2 \, (y_1)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\frac{3}{2} T_\nu^{(1)} \, (y_1)_{ij}$ | $+\frac{1}{4}(3 \left(y_2 \, (y_2)^\dagger \, y_1\right)_{ij} + 2 \left(Y_e \, (Y_e)^\dagger \, y_1\right)_{ij} + 5 \left(y_1 \, (y_1)^\dagger \, y_1\right)_{ij})$ | $-\frac{33}{4} {g_2}^2 \, (y_1)_{ij}$ |

**Quantity:** $y_1$

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{15}{4}{g_1}^2 (y_1)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{15}{4}{g_1}^2 (y_1)_{ij}$ |

$\beta_{y_2}^{(1)}$

**Quantity:** $y_2$

$$16\pi^2\,\mu\frac{dy_2}{d\mu} = \beta_{y_2}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{3}{2}T_\nu^{(2)}\,(y_2)_{ij}$ |
| T2 | $+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 5\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ |
| T3 | $-\frac{33}{4}{g_2}^2\,(y_2)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\frac{3}{2}T_\nu^{(2)}\,(y_2)_{ij}$ | $+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 5\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{33}{4}{g_2}^2\,(y_2)_{ij}$ |

**Quantity:** $y_2$

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{15}{4} {g_1}^2 (y_2)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{15}{4} {g_1}^2 (y_2)_{ij}$ |

$\beta^{(1)}_{M_F}$

**Quantity:** $M_F$

$$16\pi^2\,\mu\,\frac{dM_F}{d\mu} = \beta^{(1)}_{M_F}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{1}{2}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ |
| T2 | $-12{g_2}^2\,(M_F)_{ij}$ |
| T3 | $-6{g_1}^2\,(M_F)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\frac{1}{2}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-12{g_2}^2\,(M_F)_{ij}$ | $-6{g_1}^2\,(M_F)_{ij}$ |

$\beta^{(1)}_{m_1^2}$

**Quantity:** $m_1^2$

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu} = \beta^{(1)}_{m_1^2}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ |
| T2 | $-6\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ |
| T3 | $-\frac{9{g_2}^2 m_1^2}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ | $-6\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-\frac{9{g_2}^2 m_1^2}{2}$ |

**Quantity:** $m_1^2$

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{27{g_1}^2 m_1^2}{2}$ |
| T2 | $+4\lambda_{12}^{(1)} m_2^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{27{g_1}^2 m_1^2}{2}$ | $+4\lambda_{12}^{(1)} m_2^2$ |

$\beta^{(1)}_{m_2^2}$

**Quantity:** $m_2^2$

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu} = \beta^{(1)}_{m_2^2}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ |
| T2 | $-6\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$ |
| T3 | $-\frac{9{g_2}^2 m_2^2}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ | $-6\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$ | $-\frac{9{g_2}^2 m_2^2}{2}$ |

**Quantity:** $m_2^2$

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{3{g_1}^2 m_2^2}{2}$ |
| T2 | $+4\lambda_{12}^{(1)} m_1^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{3{g_1}^2 m_2^2}{2}$ | $+4\lambda_{12}^{(1)} m_1^2$ |

$\beta^{(1)}_{\lambda_1}$

**Quantity:** $\lambda_1$

$$16\pi^2\,\mu\frac{d\lambda_1}{d\mu} = \beta^{(1)}_{\lambda_1}$$

# Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_1 (12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ |
| T3 | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda_1 (12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |

**Quantity:** $\lambda_1$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ |
| T2 | $+4\lambda_5\ (\lambda_5)^*$ |
| T3 | $+\frac{9{g_2}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |

**Quantity:** $\lambda_1$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(\frac{{3g_2}^2}{2}-3\lambda_1\right)$ |
| T2 | $-9{g_2}^2\lambda_1$ |
| T3 | $+\frac{3{g_1}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+{g_1}^2\left(\frac{{3g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |

**Quantity:** $\lambda_1$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+12{\lambda_1}^2$ |
| T2 | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3 | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+12{\lambda_1}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ |

**Quantity:** $\lambda_1$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{HS_2}^{(1)\ 2}$ |
| T2 | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4\lambda_{HS_2}^{(1)\ 2}$ | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ |

$\beta^{(1)}_{\lambda^{(1)}_{S_1}}$

**Quantity:** $\lambda^{(1)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_1}}$$

# Terms 1 of 4

| Column | Representative term structure |
|---|---|
| T1 | $6\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ |
| T2 | $-5\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$ |
| T3 | $+\frac{9{g_2}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $6\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-5\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$ | $+\frac{9{g_2}^4}{4}$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 2 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2 \left( \frac{27{g_2}^2}{2} - 27\lambda_{S_1}^{(1)} \right)$ |
| T2 | $-9{g_2}^2\lambda_{S_1}^{(1)}$ |
| T3 | $+\frac{243{g_1}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+{g_1}^2 \left( \frac{27{g_2}^2}{2} - 27\lambda_{S_1}^{(1)} \right)$ | $-9{g_2}^2\lambda_{S_1}^{(1)}$ | $+\frac{243{g_1}^4}{4}$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 3 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{12}^{(1)\,2}$ |
| T2 | $+\frac{\lambda_{12}^{(A)\,2}}{4}$ |
| T3 | $+4\lambda_{HS_1}^{(1)\,\,2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4\lambda_{12}^{(1)\,2}$ | $+\frac{\lambda_{12}^{(A)\,2}}{4}$ | $+4\lambda_{HS_1}^{(1)\,\,2}$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 4 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ |
| T2 | $+12{\lambda_{S_1}^{(1)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ | $+12{\lambda_{S_1}^{(1)}}^2$ |

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

**Quantity:** $\lambda^{(1)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_2}}$$

## Terms 1 of 4

| Column | Representative term structure |
|---|---|
| T1 | $6\lambda^{(1)}_{S_2}T^{(2)}_{\nu}$ |
| T2 | $-5\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3 | $+\frac{9{g_2}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $6\lambda^{(1)}_{S_2}T^{(2)}_{\nu}$ | $-5\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+\frac{9{g_2}^4}{4}$ |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 2 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$ |
| T2 | $-9{g_2}^2\lambda_{S_2}^{(1)}$ |
| T3 | $+\frac{3{g_1}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$ | $-9{g_2}^2\lambda_{S_2}^{(1)}$ | $+\frac{3{g_1}^4}{4}$ |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 3 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{12}^{(1)\,2}$ |
| T2 | $+\frac{\lambda_{12}^{(A)\,2}}{4}$ |
| T3 | $+4\lambda_{HS_2}^{(1)\;\;2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4\lambda_{12}^{(1)\,2}$ | $+\frac{\lambda_{12}^{(A)\,2}}{4}$ | $+4\lambda_{HS_2}^{(1)\;\;2}$ |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 4 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T2 | $+12{\lambda_{S_2}^{(1)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$ | $+12{\lambda_{S_2}^{(1)}}^2$ |

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

**Quantity:** $\lambda^{(1)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

# Terms 1 of 4

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{HS_1}(3T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3 | $+6\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda^{(1)}_{HS_1}(3T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\,(\lambda_5)^*$ |

**Quantity:** $\lambda_{HS_1}^{(1)}$

# Terms 2 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{9{g_2}^4}{4}$ |
| T2 | $-9{g_2}^2\lambda_{HS_1}^{(1)}$ |
| T3 | $+\frac{27{g_1}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda_{HS_1}^{(1)}$ | $+\frac{27{g_1}^4}{4}$ |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 3 of 4

| Column | Representative term structure |
|---|---|
| T1 | $-15{g_1}^2\lambda^{(1)}_{HS_1}$ |
| T2 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ |
| T3 | $+6\lambda_1\lambda^{(1)}_{HS_1}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-15{g_1}^2\lambda^{(1)}_{HS_1}$ | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 4 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+4{\lambda^{(1)}_{HS_1}}^{2}$ |
| T2 | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ |
| T3 | $+\frac{3{\lambda^{(A)}_{HS_1}}^{2}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4{\lambda^{(1)}_{HS_1}}^{2}$ | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | $+\frac{3{\lambda^{(A)}_{HS_1}}^{2}}{4}$ |

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

**Quantity:** $\lambda^{(1)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

# Terms 1 of 4

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{HS_2}(3T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $-3\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3 | $+6\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda^{(1)}_{HS_2}(3T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-3\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\,(\lambda_5)^*$ |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 2 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{9{g_2}^4}{4}$ |
| T2 | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ |
| T3 | $+\frac{3{g_1}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ | $+\frac{3{g_1}^4}{4}$ |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 3 of 4

| Column | Representative term structure |
|---|---|
| T1 | $-3{g_1}^2\lambda^{(1)}_{HS_2}$ |
| T2 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ |
| T3 | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-3{g_1}^2\lambda^{(1)}_{HS_2}$ | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |

**Quantity:** $\lambda_{HS_2}^{(1)}$

# Terms 4 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T2 | $+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3 | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4{\lambda_{HS_2}^{(1)}}^2$ | $+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$ | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |

$\beta^{(1)}_{\lambda^{(1)}_{12}}$

**Quantity:** $\lambda^{(1)}_{12}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{12}}$$

## Terms 1 of 4

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{12}(3T^{(1)}_{\nu} + 3T^{(2)}_{\nu})$ |
| T2 | $-\frac{9}{2}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3 | $+3\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda^{(1)}_{12}(3T^{(1)}_{\nu} + 3T^{(2)}_{\nu})$ | $-\frac{9}{2}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+3\lambda_5\,(\lambda_5)^*$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 2 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{9{g_2}^4}{4}$ |
| T2 | $-9{g_2}^2\lambda_{12}^{(1)}$ |
| T3 | $+\frac{27{g_1}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda_{12}^{(1)}$ | $+\frac{27{g_1}^4}{4}$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 3 of 4

| Column | Representative term structure |
|---|---|
| T1 | $-15{g_1}^2\lambda_{12}^{(1)}$ |
| T2 | $+4{\lambda_{12}^{(1)}}^2$ |
| T3 | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-15{g_1}^2\lambda_{12}^{(1)}$ | $+4{\lambda_{12}^{(1)}}^2$ | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 4 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T2 | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ |
| T3 | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

$\beta^{(1)}_{\lambda_5}$

**Quantity:** $\lambda_5$

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu} = \beta^{(1)}_{\lambda_5}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_5\left(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$ |
| T2 | $-9{g_2}^2\lambda_5$ |
| T3 | $-9{g_1}^2\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda_5\left(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$ | $-9{g_2}^2\lambda_5$ | $-9{g_1}^2\lambda_5$ |

**Quantity:** $\lambda_5$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_{12}^{(1)}\lambda_5$ |
| T2 | $+2\lambda_1\lambda_5$ |
| T3 | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

**Quantity:** $\lambda_5$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{HS_2}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ |

$\beta^{(1)}_{\lambda_{12}^{(A)}}$

**Quantity:** $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(A)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(A)}}$$

## Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$ |
| T2 | $-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3 | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$ | $-\frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$ |
| T2 | $-9{g_2}^2\lambda_{12}^{(A)}$ |
| T3 | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$ | $-9{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ |
| T2 | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ |
| T3 | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

**Quantity:** $\lambda^{(A)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(A)}_{HS_1}(3T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3 | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda^{(A)}_{HS_1}(3T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |

**Quantity:** $\lambda_{HS_1}^{(A)}$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(-18{g_2}^2-15\lambda_{HS_1}^{(A)}\right)$ |
| T2 | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ |
| T3 | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+{g_1}^2\left(-18{g_2}^2-15\lambda_{HS_1}^{(A)}\right)$ | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |

**Quantity:** $\lambda^{(A)}_{HS_1}$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_1\lambda^{(A)}_{HS_1}$ |
| T2 | $+8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$ |
| T3 | $+2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+2\lambda_1\lambda^{(A)}_{HS_1}$ | $+8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$ | $+2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$ |

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

**Quantity:** $\lambda^{(A)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_2}}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(A)}_{HS_2}(3T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3 | $+16\lambda_5\ (\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda^{(A)}_{HS_2}(3T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+16\lambda_5\ (\lambda_5)^*$ |

**Quantity:** $\lambda_{HS_2}^{(A)}$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$ |
| T2 | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ |
| T3 | $+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$ | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ |

**Quantity:** $\lambda_{HS_2}^{(A)}$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T2 | $+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$ |
| T3 | $+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+2\lambda_1\lambda_{HS_2}^{(A)}$ | $+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$ | $+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$ |